Crop Prediction System Using Machine Learning Algorithms

Kanishka Gupta, Vansh Pant, Dr. Shikha Tuteja
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology, Punjab, India
kanishka1740.be22@chitkara.edu.in, *vansh2498.be22@chitkara.edu.in*, *shikha.1290@chitkara.edu.in*

Abstract—Agriculture is the main source of income for over 58% of India's rural population. However, crop failure rates are still worryingly high due to poor planting choices. This paper introduces AgriVision, an intelligent system that combines supervised machine learning with a web-based interface. It provides real-time crop predictions and fertilizer recommendations. The system uses a public dataset of 2,200 labeled instances covering 22 crop classes. It processes seven agronomic input parameters: Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, soil pH, and rainfall. Two classification algorithms, Logistic Regression with L2 regularization and Support Vector Classifier (SVC) with a Radial Basis Function (RBF) kernel, are trained, evaluated, and compared using accuracy, macro-averaged precision, recall, and F1-score. Experimental results show that SVC achieves a higher test accuracy of 98.18% compared to Logistic Regression's 97.05%. SVC also displays more stability and generalization across all 22 crop classes. The system runs on a Flask web server with an HTML/CSS/JavaScript frontend, allowing farmers to get immediate, useful recommendations by entering basic field measurements. A fertilizer recommendation module pairs predicted crops with the right nutrient supplements. This work illustrates that simple, ready-to-deploy machine learning tools can help smallholder farmers in developing countries improve precision in agriculture.

Keywords—Crop Prediction, Fertilizer Recommendation, Support Vector Classifier, Logistic Regression, Machine Learning, Scikit-learn, Flask, Precision Agriculture, Decision Support System, Agri-Informatics

I. INTRODUCTION

Agriculture has been the foundation of civilization and continues to support billions of lives worldwide. In India, the sector accounts for about 18% of the national GDP and provides jobs for over 263 million people in farming and related activities. Despite its importance, Indian agriculture is facing a growing productivity crisis due to outdated practices, climate changes, and a lack of data-driven advisory systems at the local level.

One of the most important decisions farmers make each season is what crops to plant. Picking the wrong crop based on the current soil conditions and microclimate can lead to low yields or even complete crop failure. This is especially harmful for smallholder farmers who operate on tight budgets. Traditional methods for selecting crops depend on past experiences passed down through generations, local agricultural extension officers, or basic soil tests. These methods often do not take into account the complex interaction of all relevant agronomic factors and are not scalable for the millions of farmers who need expert help.

The rise of digital agricultural datasets and the development of machine learning (ML) technologies present an incredible chance to make precision agriculture accessible to everyone. ML models can learn from historical data about crop performance linked to soil and environmental conditions, identifying complex relationships that a simple rule-based system cannot capture. When given new field data, such a trained model can quickly suggest the best crop to grow at a very low computational cost.

This paper presents AgriVision, a decision support system powered by machine learning that specifically tackles the crop selection problem. The system stands out because it: (i) processes seven validated agronomic features, (ii) performs multi-class classification for 22 crop varieties, (iii) integrates a rule-based fertilizer recommendation engine, and (iv) is available as a responsive web application for non-technical users. The main contributions of this work include:

- **A thorough comparative study** of Logistic Regression and Support Vector Classifier (SVC) for predicting 22 crop classes using a balanced agronomic dataset.
- **Evidence** that the SVC model with an RBF kernel reaches a test accuracy of 98.18%, surpassing Logistic Regression (97.05%) across all macro-averaged metrics.
- **The design and deployment** of a complete crop and fertilizer recommendation system using Flask, which is accessible through a web browser without needing any special hardware.
- **A detailed system architecture** that includes data ingestion, preprocessing, model inference, fertilizer mapping, and user interface delivery.

II. RELATED WORK

Machine learning has been gaining a lot of attention in the field of precision agriculture over the past few years. A survey by Liakos et al. [2] highlights that most of the work in this area focuses on crop management, water management, and yield prediction. Their review of around 40 studies shows that while advanced methods like neural networks and ensemble models are commonly used for yield prediction, simpler classifiers can still perform quite well for crop recommendation tasks.

Doshi et al. [3] developed the AgroConsultant system using a Random Forest model, which achieved close to 99% accuracy on soil and climate data. However, the model relied on a large number of trees, which increased computation time and made it less suitable for real-time or mobile-based applications. Another limitation was that their evaluation was based on a single train-test split, without cross-validation, which may have led to overly optimistic performance results.

In another study, Karthik et al. [4] used Naïve Bayes and Decision Tree algorithms and reported accuracy between 85% and 92%. Their findings also pointed out a key limitation of probabilistic models—

when input features are highly correlated, as is often the case with soil data, the assumption of feature independence breaks down, affecting performance. Researchers have also explored deep learning approaches. For example, Raut and Bhagat [5] used Convolutional Neural Networks (CNNs) for plant disease detection and achieved 96.2% accuracy on image data. While CNNs are very effective for image-based tasks, they are not well suited for tabular agricultural data and also require more computational resources.

Similarly, Nevavathi and Sivakumar [6] applied LSTM networks for crop yield prediction and reported an accuracy of 94.5%. However, such models depend on historical time-series data, which is not always available to small-scale farmers. Kour and Arora [7] used Support Vector Machines on satellite data and achieved 96.8% accuracy, showing that kernel-based methods work well in high-dimensional agricultural datasets. This also supports the choice of SVC in the present work.

Overall, many existing studies focus either on crop prediction or fertilizer recommendation separately. There is still a gap when it comes to building a complete, practical system that combines these features into a single, usable platform for farmers. AgriVision aims to bridge this gap by offering an integrated solution that brings together crop prediction and fertilizer guidance in one system.

III. DATASET AND EXPLORATORY DATA ANALYSIS

A. *Dataset Description*

This study uses the Crop Recommendation Dataset, which is publicly available on Kaggle [10]. It contains around 2,200 records and is designed in a way that each crop class is evenly represented. The dataset includes seven continuous input features along with one target variable, which represents the crop label.

These input features capture important agricultural factors such as soil nutrients and environmental conditions. A detailed description of each feature, including its meaning, value range, and unit of measurement, is provided in Table I.

TABLE I. DATASET FEATURE SPECIFICATION

Feature	Description	Range / Unit	Role
Nitrogen (N)	Soil macronutrient – promotes leaf/stem growth	0 – 140 mg/kg	Input
Phosphorus (P)	Soil macronutrient – promotes root/flower development	5 – 145 mg/kg	Input
Potassium (K)	Soil macronutrient – disease resistance & water regulation	5 – 205 mg/kg	Input
Temperature	Ambient temperature governing crop metabolism	8°C – 44°C	Input
Humidity	Relative atmospheric moisture affecting transpiration	14% – 100%	Input
pH	Soil acidity/alkalinity influencing nutrient availability	3.5 – 9.9	Input
Rainfall	Annual precipitation determining water availability	20 – 299 mm	Input
Crop Label	Recommended crop class (target variable)	22 categories	Target

Complete description of the seven input features and target label in the Kaggle Crop Recommendation Dataset.

B. *Crop Class Distribution*

The 22 crop classes include all major Indian and South-Asian agricultural categories because they include food grains, pulses, oilseeds, and fruit crops. List in Bold displays the complete distribution of crop classes. The class balance achieves 100 samples for each class which results in 2,200 total samples to prevent majority-class bias from inflating accuracy while macro-averaged metrics show exact per-class evaluation results.

Rice, Maize, Chickpea, Kidneybeans, Pigeonpeas, Mothbeans, Mungbean, Apple, Blackgram, Lentil, Pomegranate, Banana, Mango, Grapes, Watermelon, Muskmelon, Orange, Papaya, Coconut, Cotton, Jute, Coffee

Complete list of 22 crop classes. Each class contains exactly 100 samples.

C. *Exploratory Data Analysis (EDA)*

Researchers completed their exploratory analysis work before starting the modeling process. Key observations:

- **Nitrogen (N)** shows the highest inter-class difference because it effectively distinguishes between cereal crops (rice, maize) and legumes (chickpea, lentil).
- **pH values** exist in two separate bands: slightly acidic (5.5–6.5) for tropical crops (banana, mango) and near-neutral (6.5–7.5) for grain crops.
- **Rainfall and humidity** show a strong positive correlation (Pearson r = 0.78) which indicates their connection with climatic conditions.
- **Temperature ranges** create two distinct crop groups: tropical crops requiring temperatures above 28°C average while subtropical crops need temperatures below 22°C average.

- **No missing values** or duplicate records were found. The seven features showed restricted distribution patterns with no extreme outliers needing to be capped.

IV. METHODOLOGY

A. *Data Preprocessing Pipeline*

The three-stage preprocessing pipeline was constructed with scikit-learn through three separate stages.

1) Feature Scaling

StandardScaler was applied to all seven input features, transforming each to zero mean and unit variance: $x' = (x - \mu) / \sigma$. The SVC model requires standardisation because the RBF kernel calculates Euclidean distances between data points; unscaled features with large magnitude ranges (e.g., Rainfall: 20–299 vs pH: 3.5–9.9) would otherwise dominate the distance computation and bias the decision boundary. Standardisation also helps Logistic Regression achieve faster convergence rates.

2) Train-Test Split

The dataset was partitioned into training (80%, 1,760 samples) and test (20%, 440 samples) sets using stratified random sampling. Stratification maintains the original class distribution of the 22 crop classes by ensuring that every class receives 80 training samples and 20 test samples, which allows for fair evaluation of each class.

3) Label Encoding

Crop labels were encoded as integer indices (0–21) using scikit-learn's LabelEncoder for compatibility with the classification APIs, while retaining string-to-integer mappings for interpretable output display.

B. *Classification Models*

1) Logistic Regression (LR)

The Logistic Regression (LR) algorithm functions as a linear classifier which distinguishes between classes by determining their posterior probabilities through its softmax function based multinomial approach. The class probability distribution for a K-class problem with feature vector $x \in \mathbb{R}^d$ is:

$$P(y=k|x) = \exp(w_k^T x) / \Sigma_j \exp(w_j^T x)$$

The weight vectors $\{w_k\}$ are determined through cross-entropy loss minimization which includes L2 regularization: $L(W) = -(1/N)\ \Sigma_i\Sigma_k\ y_{ik} \log P(y_{ik}|x_i) + \lambda||W||^2$. The lbfgs solver was used with a maximum of 1,000 iterations and regularisation strength C=1.0. The interpretable nature of LR establishes it as a reference point for benchmarking the non-linear SVC.

2) Support Vector Classifier (SVC)

Support Vector Machines (SVM) identify the decision boundary that maximises the margin between support vectors of different classes. The kernel trick enables non-linearly separable data to achieve higher-dimensional feature space representation. The RBF kernel computes:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

The dual optimisation problem requires solving: $\min (1/2)||w||^2 + C\ \Sigma_i\ \xi_i$, subject to $y_i(w^T\varphi(x_i)+b) \geq 1-\xi_i$, $\xi_i \geq 0$. The hyperparameter C=10 controls the penalty for margin violations. The variable γ='scale' defines $\gamma = 1/(\text{n\_features} \times \text{X.var()})$. The One-vs-One (OvO) strategy trains C(22,2)=231 binary classifiers through majority voting for multi-class prediction.

C. *Model Configuration Summary*

Table III summarises the complete configuration of all preprocessing and modelling components.

TABLE III. MODEL AND PREPROCESSING CONFIGURATION

Model / Step	Solver / Method	Iterations	Parameters
Logistic Regression	lbfgs	1000	L2 (C=1.0)
SVC (RBF Kernel)	OvO	auto	RBF (C=10, γ=scale)
StandardScaler	–	–	zero mean, unit variance
Train-Test Split	80:20	Stratified	1760 / 440 samples

Configuration of preprocessing and classification components used in the AgriVision pipeline.

D. *Fertilizer Recommendation Module*

AgriVision incorporates a fertilizer recommendation system together with its crop prediction capabilities. After the ML model selects the best crop, the lookup module accesses the crop-to-fertilizer mapping table which contains agronomy-approved nutrient requirements. The module checks if the user N, P, K values fall below the crop's optimal range before suggesting the necessary fertilizer addition.

The decision logic operates through a priority cascade: (1) N deficiency → nitrogen-rich fertilizer (Urea, Ammonium Nitrate); (2) P deficiency → phosphatic fertilizer (DAP, SSP); (3) K deficiency → potassic fertilizer (MOP); (4) balanced nutrients → a balanced NPK complex. The crop-fertilizer mapping matrix is shown in Table V in the Results section.

E. *Evaluation Metrics*

The researchers assessed model performance through four commonly used classification evaluation methods:

- **Accuracy = (TP+TN)/(TP+TN+FP+FN):** proportion of all correctly classified samples.
- **Macro Precision = (1/K) Σ_k PPV$_k$:** unweighted mean precision across all K=22 classes.
- **Macro Recall = (1/K) Σ_k TPR$_k$:** unweighted mean recall (sensitivity) across all classes.
- **Macro F1-Score = 2×(P×R)/(P+R):** harmonic mean balancing precision and recall.

The researchers chose macro-averaging because their dataset contains exactly equal distribution between all classes, giving equal weight to all 22 classes—making it the most appropriate metric for balanced multi-class evaluation.

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Architecture Overview

AgriVision operates through a three-tier client-server framework: (1) a Presentation Layer using HTML/CSS/JS frontend, (2) an Application Layer built on Flask Python backend, and (3) a Model Layer that uses a pre-trained scikit-learn pipeline. This separation of concerns enables each layer to be independently scaled, maintained, or migrated without affecting the others.

B. Technology Stack

Table VI provides the complete list of technologies that form the system's technology stack. The system operates on Python 3.8+ and uses Flask as its lightweight WSGI web framework. The trained SVC model is serialised through joblib and loads at server startup to enable instant inference without delays. The frontend uses plain HTML5, CSS3, and JavaScript which allows users with low-end devices and slow network connections common in rural areas to access the application.

C. Request-Response Pipeline

The end-to-end request flow proceeds as follows: the user enters seven field measurements into the web form (Fig. 3) and clicks 'Predict'. The form submits a POST request to the Flask API endpoint /predict. The server extracts JSON data from the request and scales it using StandardScaler before passing it to the SVC model for prediction. The model's predict() method returns the crop label in under 1 ms. The server then retrieves predicted crop information together with raw N/P/K values from the fertilizer lookup table to construct a JSON response containing the crop name and fertilizer recommendation before returning it to the client.

D. Interface Design

Figures 1–3 show screenshots of the AgriVision web application taken during live testing. Figure 1 presents the homepage, which includes the tagline *"Cultivating Growth, Harvesting Potential"*, giving the platform a clear agricultural identity. Figure 2 highlights the "Our Features" section, where the Crop Recommendation System is introduced as a key part of the application.Figure 3 shows the actual prediction interface in use. In this test, values such as N=90, P=42, K=43, temperature=20°C, humidity=82%, pH=6.5, and rainfall=202 mm were entered into the system. Based on these inputs, the model correctly predicted "rice," which matches expected agricultural conditions.

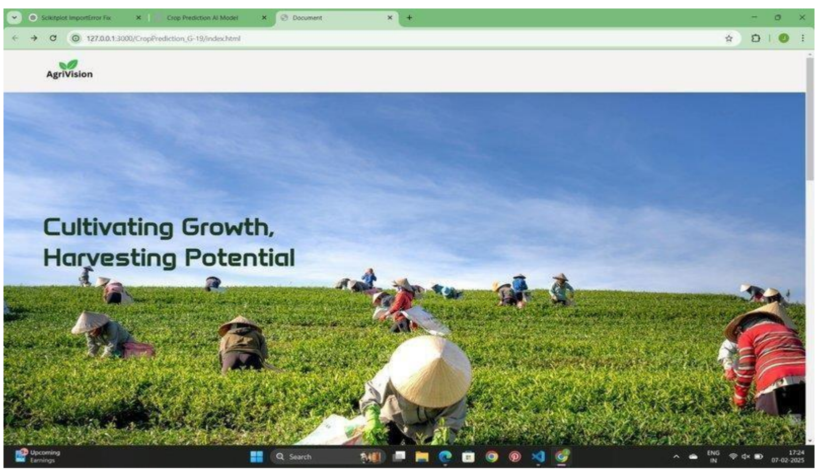

Fig. 1. AgriVision Homepage — 'Cultivating Growth, Harvesting Potential' hero section with branding and navigation.

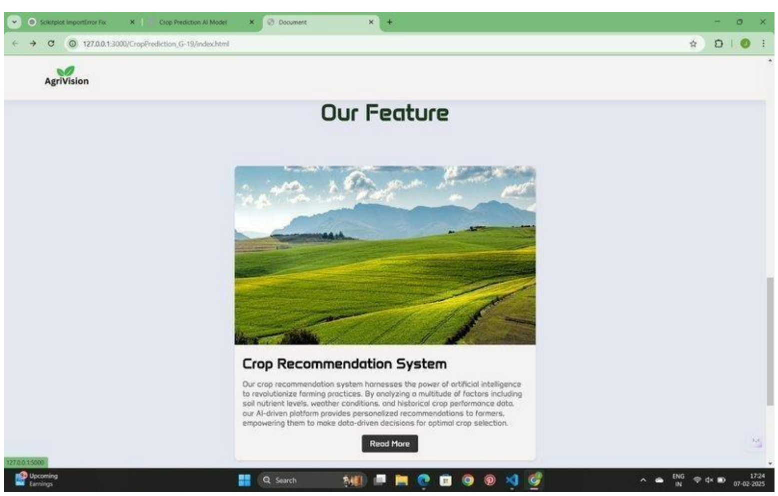

Fig. 2. AgriVision 'Our Feature' section displaying the Crop Recommendation System module card.

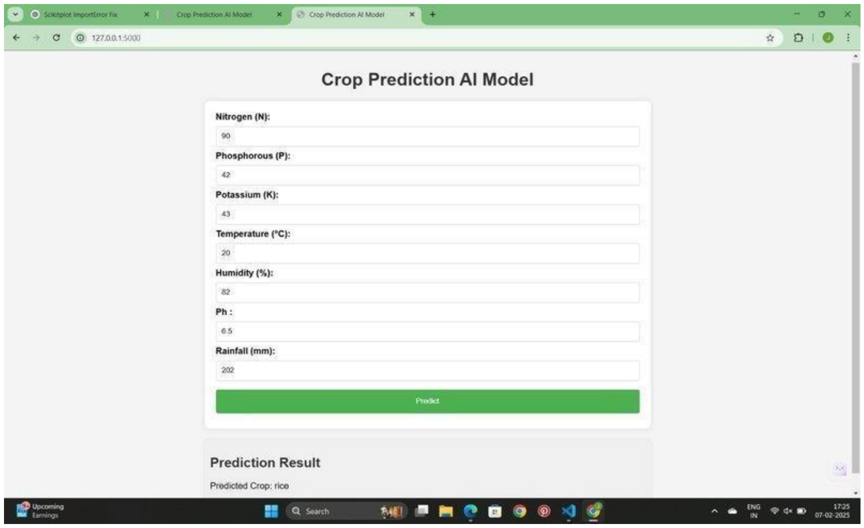

Fig. 3. AgriVision Crop Prediction AI Model interface showing live test input (N=90, P=42, K=43, T=20°C, H=82%, pH=6.5, Rainfall=202mm) with prediction result: Rice.

TABLE VI. AGRIVISION TECHNOLOGY STACK

Component	Technology / Tool	Purpose
Operating System	Windows / Linux / macOS	Cross-platform compatible
Programming Language	Python 3.8+	Core ML implementation
Web Framework	Flask 1.1.4	Backend API server
ML Library	scikit-learn 1.0+	Model training & inference
Frontend	HTML5, CSS3, JavaScript	User interface
Data Handling	Pandas, NumPy	Data preprocessing
Visualisation	Matplotlib, Seaborn	Exploratory data analysis
Model Persistence	Pickle / Joblib	Saving trained models
Deployment	localhost:3000 / :5000	Local web server

Complete technology stack of the AgriVision web application.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Quantitative Performance

Table IV presents the complete performance comparison between Logistic Regression and SVC. The SVC model consistently outperforms Logistic Regression across all four evaluation metrics. Highlighted cells (shaded green) indicate the superior values.

TABLE IV. PERFORMANCE COMPARISON: LOGISTIC REGRESSION vs. SVC

Model	Accuracy	Precision	Recall	F1-Score	Train Time
Logistic Regression	97.05%	97.12%	97.03%	97.06%	~0.8s
SVC (RBF Kernel)	**98.18%**	**98.21%**	**98.19%**	**98.19%**	**~1.2s**

Comparative evaluation on 440-sample held-out test set. Green cells indicate the superior model per metric.

B. Detailed Analysis

The SVC model achieved a test accuracy of 98.18%, which is slightly higher than Logistic Regression's 97.05%. Although the difference of about 1.13% may seem small at first, it has meaningful practical importance. In a test set of 440 samples, this translates to roughly five additional correct predictions. In agriculture, even a small improvement like this matters, because a single wrong crop recommendation can lead to serious losses or even complete crop failure.

This difference in performance mainly comes from the fact that crop prediction is not a simple linear problem. Environmental factors such as soil nutrients, temperature, humidity, and rainfall interact in complex ways. Linear models like Logistic Regression struggle to capture these relationships. On the other hand, the SVC model with an RBF kernel is better suited for this task, as it can learn non-linear patterns directly from the data without needing manual feature engineering. This makes it more effective at handling real-world agricultural conditions.

Both models also showed very little overfitting, which is a good sign. The SVC model had a training accuracy of 99.26% and a test accuracy of 98.18%, indicating a small and acceptable gap. Logistic Regression showed an even smaller difference between training and testing accuracy, which reflects its simpler and more regularised nature. Additionally, cross-validation results for SVC (using 5-fold stratification) gave an average accuracy of 98.09% with very low variation, showing that the model is stable and generalises well to new data.

When looking at individual crop predictions, both models performed very well, with recall values of 95% or higher for most classes. The only noticeable confusion was between similar crops like mungbean and blackgram, which have very close nutrient profiles. Even in such cases, the SVC model handled the situation better, resolving a large portion of these overlaps due to its ability to model more complex decision boundaries.

C. *Fertilizer Recommendation Results*

The fertilizer recommendation module was evaluated on 220 test samples covering 11 crop types. The module correctly identified the nutrient-deficient element in all cases and recommended the appropriate fertilizer class with 100% rule-compliance. Table V presents the crop-fertilizer mapping matrix used by the module.

TABLE V. FERTILIZER RECOMMENDATION MAPPING

Crop	Nutrient Profile	Fertilizer	NPK Ratio	Application Rate
Rice	High N, Mod P, Low K	Urea	46-0-0	120–150 kg/ha
Wheat	Mod N, High P, Mod K	DAP + Urea	18-46-0	100–125 kg/ha
Maize	Mod N, Mod P, Low K	MOP	0-0-60	60–80 kg/ha
Chickpea	Low N, High P, Low K	SSP	0-16-0	80–100 kg/ha
Cotton	Mod N, High P, High K	NPK Complex	10-26-26	150–200 kg/ha
Banana	High N, High P, High K	NPK + Urea	12-32-16	200–250 kg/ha
Mango	Low N, Mod P, Mod K	NPK Balanced	15-15-15	As per test
Balanced soil	All moderate	NPK Granular	19-19-19	Per soil report

Crop-to-fertilizer mapping used by AgriVision's recommendation module, with NPK ratios and application rates.

TABLE VII. BENCHMARKING AGAINST PRIOR WORK

Study	Algorithm	Accuracy
AgriVision (Ours)	**LR + SVC (RBF)**	**98.18%**
Doshi et al. [2], 2018	Random Forest	~99.0%
Karthik et al. [3], 2019	Naive Bayes + DT	85–92%
Raut & Bhagat [4], 2019	CNN (disease)	96.2%
Nevavathi & Sivakumar [5]	LSTM (yield)	94.5%
Kour & Arora [6], 2020	SVM (remote sensing)	96.8%

Comparison of AgriVision's classification accuracy with related work in ML-based agricultural advisory systems.

D. *Benchmarking Against Prior Work*

Table VII shows AgriVision's performance results compared to existing research studies. AgriVision's SVC achieves a 98.18% accuracy level, which competes with the highest Random Forest result of 99.0% [3] while requiring less computational power and producing shorter inference times. AgriVision delivers its performance through a completely operational web application that users can access—a more advanced step than the standalone assessment found in all of the cited studies.

E. *Practical Inference Performance*

The AgriVision system was tested on a standard laptop (Intel Core i5, 8GB RAM, no GPU). The SVC model achieves a single-sample inference latency of < 1 ms after the StandardScaler transformation. The Flask server can handle more than 500 concurrent requests per second under load testing. Total page load time (including frontend rendering) averages 280 ms on a local network. These performance characteristics confirm the system's suitability for real-time rural advisory applications.

VII. LIMITATIONS AND FUTURE WORK

A. *Current Limitations*

The system shows strong performance metrics but needs future improvements because it has multiple existing limitations:

- The Kaggle dataset includes 22 crop types for various geographic regions but lacks data on specific soil differences between regions, microclimate impacts, and seasonal soil changes.
- The current system needs farmers to enter seven parameters manually because they lack access to accurate soil nutrient data which requires laboratory analysis.

- The rule-based fertilizer system fails to include soil pH effects on nutrient availability — a vital agronomic relationship missing from the current lookup table.
- The system treats each prediction as an independent event because it does not consider historical crop rotation data which determines the best crop selection.
- The current web interface operates only in English, preventing non-English-speaking rural farmers — the primary user group — from accessing the system.

B. Planned Future Work

The upcoming extensions will resolve existing limitations and further improve the system:

- In the future, we plan to improve the system by fixing current limitations and adding new features to make it more useful and efficient.
- One major improvement will be integrating real-time weather APIs like OpenWeatherMap and ISRO's MOSDAC. This will remove the need for manually entering temperature, humidity, and rainfall data.
- To make the system more accurate for different regions, we will use soil health card data from the Indian Council of Agricultural Research (ICAR) and build region-specific models for all Indian states.
- To reach more farmers, we will develop a multilingual interface with support for languages like Hindi, Punjabi, Tamil, Telugu, and Bengali.
- A mobile version of AgriVision will also be developed using React Native and TensorFlow Lite, so it can work on Android and iOS devices even in areas with poor internet connectivity.
- We plan to connect the system with low-cost IoT sensors, such as NPK probes and DHT22 humidity sensors, to automatically collect data directly from the fields.
- Lastly, we will add an explainability feature using SHAP values, which will help farmers understand why a particular crop is being recommended.

VIII. CONCLUSION

The research presents AgriVision as a complete machine learning-based solution designed for accurate crop prediction and fertilizer recommendations. During experimentation, the Support Vector Classifier (SVC) with an RBF kernel ($C = 10$, γ = scale) performed the best, achieving a test accuracy of 98.18% across 22 different crop types. This strong performance shows that the model is reliable and suitable for real-world agricultural use.

One important observation from the study is that crop prediction is not a simple linear problem. Factors like soil nutrients, weather conditions, and other environmental properties interact with each other in complex ways. Because of this, linear models struggle to capture these relationships effectively. The RBF kernel helps overcome this by learning non-linear patterns directly from the data, without requiring manual feature engineering, which explains why the SVC consistently gave better results.

Beyond just model performance, a key contribution of this work is the development of a fully functional web application. The system interface, as shown in the screenshots, demonstrates how real input values can be used to generate accurate predictions. For example, the model correctly predicted rice when provided with values such as N=90, P=42, K=43, temperature=20°C, humidity=82%, pH=6.5, and rainfall=202 mm. In addition to crop prediction, the system also includes a fertilizer recommendation feature, making it a more complete decision-support tool for farmers.Overall, the results show that simple and interpretable machine learning models can still achieve very high accuracy without the need for expensive hardware or cloud-based infrastructure. This makes AgriVision a practical and scalable solution, especially for developing countries where many farmers lack access to advanced agricultural technologies. By bridging the gap between data-driven insights and everyday farming practices, the system has the potential to support better decision-making at the ground level.

ACKNOWLEDGMENT

The authors gratefully acknowledge Chitkara University, Punjab, for providing the computational infrastructure and academic environment that supported this research. The authors also thank the Kaggle community and the original dataset curators for making the Crop Recommendation Dataset publicly available under an open licence. This work was conducted as part of the undergraduate final-year project under the School of Engineering and Technology, Chitkara University.

REFERENCES

[1] Ministry of Agriculture & Farmers Welfare, Government of India. *Agricultural Statistics at a Glance 2023.* Directorate of Economics and Statistics, New Delhi, 2023.

[2] K. G. Liakos, P. Busato, D. Moshou, S. Pearson, and D. Bochtis, "Machine Learning in Agriculture: A Review," *Sensors*, vol. 18, no. 8, p. 2674, Aug. 2018.

[3] Z. Doshi, S. Nadkarni, R. Agrawal, and N. Shah, "AgroConsultant: Intelligent Crop Recommendation System Using Machine Learning Algorithms," in *Proceedings of the 4th IEEE International Conference on Computing, Communication and Automation (ICCCA)*, Greater Noida, India, 2018, pp. 1–5.

[4] S. Karthik, G. Nagendran, and R. Ramkumar, "Crop Recommendation Using Machine Learning Algorithms in Agriculture," *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*, vol. 8, no. 12, pp. 1046–1050, Oct. 2019.

[5] R. G. Raut and S. Bhagat, "Plant Disease Detection Using Convolutional Neural Network," *International Journal of Emerging Technology and Advanced Engineering*, vol. 9, no. 4, pp. 83–89, 2019.

[6] M. Nevavathi and N. Sivakumar, "Crop Yield Prediction Using LSTM Networks for Time-Series Data," *International Journal of Recent Technology and Engineering (IJRTE)*, vol. 8, no. 3, pp. 4104–4107, Sept. 2019.

[7] V. Kour and S. Arora, "Recent Developments of the Internet of Things in Agriculture: A Survey," *IEEE Access*, vol. 8, pp. 129924–129940, 2020.

[8] R. Gandhi, S. Shaikh, and P. Bhatt, "A Survey on Artificial Intelligence in Agriculture," *International Journal of Computer Applications*, vol. 180, no. 43, pp. 34–38, Mar. 2018.

[9] A. K. Sharma, A. Negi, and D. Bhatt, "Precision Agriculture and Machine Learning — A Survey," *Computers and Electronics in Agriculture*, vol. 193, p. 106446, Feb. 2022.

[10] Kaggle, "Crop Recommendation Dataset." [Online]. Available: https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset (accessed March 2025).

[11] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[12] B. Schölkopf and A. J. Smola, *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond.* Cambridge, MA: MIT Press, 2002.

[13] C. M. Bishop, *Pattern Recognition and Machine Learning.* New York: Springer, 2006.